Lectures 7 & 8

- finish remaining lecture 6 material (abbreviated version here)
- expected utility: theory and assumptions
- risk attitudes
- EU applications

Example From Last Class

- assume $u(x_1, x_2) = x_1 x_2$
- calculate the compensated demand and expenditure functions
- then, assume $m = 10$, $p_2 = 1$, and p_1 increases from \$1 to \$4
- find the substitution effect, income effect, and CV

Recap of Solution So Far

- last class, we learned that the compensated demand bundle is

$$(x_1^c, x_2^c)(p_1, p_2, u^*) = \left(\sqrt{\frac{p_2}{p_1}u^*}, \sqrt{\frac{p_1}{p_2}u^*}\right)$$

  - this is the bundle that most cheaply yields utility u^*, given prices p_1 and p_2
  - to get it, we solved $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ (here $\frac{x_2}{x_1} = \frac{p_1}{p_2}$) and $u(x_1, x_2) = u^*$ (here $x_1 x_2 = u^*$)
- and the expenditure function is the cost of this bundle, generally $e(p_1, p_2, u^*) = p_1 x_1^c + p_2 x_2^c$, here

$$e(p_1, p_2, u^*) = p_1\sqrt{\frac{p_2}{p_1}u^*} + p_2\sqrt{\frac{p_1}{p_2}u^*} = 2\sqrt{p_1 p_2 u^*}$$

  - this is how much money you need to get utility u^*, given prices p_1 and p_2

Application Part: Overview

Now for income and substitution effects and CV, at income m if p_1 rises to p_1',

- Step 1: find original bundle x^o and original utility u^o
- Step 2: find bundle x^c that's optimized at new prices, but exactly as good as x^o
  - direct method: solve $u(x_1^c, x_2^c) = u(x_1^o, x_2^o)$ and $\frac{MU_1}{MU_2} = \frac{p_1'}{p_2}$ (or appropriate caveat)
  - or, if you already have compensated demand, this is $(x_1^c, x_2^c)(p_1', p_2, u^o)$
- Step 3: find bundle x^f that's optimized at new prices but affordable with income m
- income effect $= x_1^f - x_1^c$, substitution effect $= x_1^c - x_1^o$
- CV $=$ subsidy needed to afford the bundle x^c in Step 2 (cost of this bundle at new prices, minus m)
  - alternatively, it is $e(p_1', p_2, u^o) - m$: cost of getting old utility at new prices, minus its old cost m

Application Part: Solution to Example

- Step 1: these are Cobb Douglas preferences with equal exponents, so spend $\frac{m}{2} = \$5$ on each good
  - at original prices $p_1 = p_2 = 1$, this lets you buy $(x_1^o, x_2^o) = (5, 5)$ for utility $u^o = 5 \cdot 5 = 25$
- Step 2: we already have compensated demand (2 slides up), so evaluating at new prices and old utility,

$$(x_1^c, x_2^c)(p_1', p_2, u^o) = \left[\left(\sqrt{\frac{p_2}{p_1}u^*}, \sqrt{\frac{p_1}{p_2}u^*}\right)\right]_{p_1=4, p_2=1, u^*=25} = \left(\frac{5}{2}, 10\right)$$

- Step 3: again since these are Cobb Douglas, at the end of the day spend $\frac{m}{2} = \$5$ on each good
  - at new prices $p_1' = 4$ and $p_2 = 1$, this gets you $(x_1^f, x_2^f) = \left(\frac{5}{4}, 5\right)$
- income effect $\frac{5}{4} - \frac{5}{2} = -1.25$ and subsititution effect $\frac{5}{2} - 5 = -2.5$
- CV: the Step 2 bundle at new prices costs $\left(\frac{5}{2}, 10\right) \cdot (\$4, \$1) = \$20$ (or use $e(4, 1, 25) = 2\sqrt{4 \cdot 1 \cdot 25} = \$20$)

Application Part: Wrap-Up

- ultimately, when p_1 increases from \$1 to \$4, the consumer reduces good 1 consumption by 3.75 units (5 to 1.25)
  - 2.5 units of this is due to reoptimization: even if a subsidy eliminated wealth effects, there's a smarter way to allocate resources at the new prices, namely shift some money to good 2
  - 1.25 units of this is due to wealth effects: the consumer is worse off with the high prices and corresponding fall in spending power
- it would take a \$10 subsidy (the CV) for the consumer to be unharmed by the price change: in essence, CV measures the welfare cost of the price increase in dollar terms
- notice that at the new prices, the original bundle (5,5) costs $(5{,}5)\cdot(\$4,1) = \$25$, which is \$15 more than wealth m
  - it would take an extra \$15 to exactly buy the original bundle
  - but the consumer would only need \$10 to maintain utility, by switching to the bundle x^c that's just as good but more smartly allocates money at the new prices!

Shephard's Lemma

The expenditure function and Hicksian demand functions satisfy the following relationship:

$$\frac{\partial e(p,u)}{\partial p_i} = x_i^c(p,u)$$

- earlier we learned that if you already know $x_1^c(p,u^*)$ and $x_2^c(p,u^*)$, calculate their cost to get the expenditure function $e(p,u)$
- this says that conversely if you know the expenditure function, you can easily get the x_i^c's

Explaining Shephard's Lemma

- recall:

$$e(p_1, p_2, u^*) = p_1 x_1^c(p_1, p_2, u^*) + p_2 x_2^c(p_1, p_2, u^*)$$

- if you differentiate this in p_1, you get direct effect $x_1^c(p_1, p_2, u^*)$ plus indirect effect $p_1 \frac{\partial x_1^c}{\partial p_1} + p_2 \frac{\partial x_2^c}{\partial p_1}$
- but the *envelope theorem* says that by virtue of the optimization process, when differentiating a value function in a parameter, only direct effects matter

Envelope Theorem: Illustration in Single-Variable Case

- suppose you choose x to minimize (or maximize) a function $f(x, p)$, where p is a parameter
- and you get an optimal solution $x^*(p)$, solving FOC $f_x(x^*(p), p) = 0$
- then $\frac{d}{dp} f\left(x^*(p), p\right) = \underbrace{f_x \cdot \dfrac{d}{dp} x^*(p)}_{\text{indirect effect}} + f_p$
- but $f_x = 0$ from the FOC, so the indirect effect vanishes!
- envelope theorem says this holds with multiple variables as well

New Topic: Expected Utility

- the next few classes will look at choices under uncertainty
- for example, should you carry an umbrella?
  - depends on the chance of rain!
- how should you allocate your money across several risky assets?

Motivation: St Petersburg Paradox

- how much would you pay to play a game that pays \$2 with chance $\frac{1}{2}$, \$4 with chance $\frac{1}{4}$, ..., $\$2^n$ with chance $\frac{1}{2^n}$?
- if all people cared about was *expected monetary value*, they should be willing to pay ∞:
  - you win $\$2^1$ with chance $\frac{1}{2^1}$, $\$2^2$ with chance $\frac{1}{2^2}$, and so on
  - so the EXPECTED prize is $\sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 = \infty$
- however, I doubt anyone would be willing to pay ∞ to pay it!
- why not?

Answer: Risk

- while the expected amount of money you win is infinite, there is a possibility that you will end up with just \$2, or just \$4, etc.
- a risk-averse person dislikes this risk, and must be compensated for it: the gamble is worth less to him than its expected monetary payoff
- we'll capture this by assuming that people care about *expected utility*, not expected wealth

Expected Utility: Calculation

- in the case where outcomes are monetary prizes, assume a utility-of-money function, $u(w)$
- *expected utility* simply calculates the expected value of final utility
- if the final wealth level could be w_1, or w_2, or.... or w_n, with probabilities $p_1, p_2, ..., p_n$, then the *expected utility* is:

$$EU \equiv E[u(w)] = p_1 u(w_1) + p_2 u(w_2) + ... + p_n u(w_n)$$

Example

- consider a guy with initial wealth \$200, who gets utility $u(w) = \sqrt{w}$ from wealth w and faces the following gamble:
  - with chance $\frac{1}{3}$, gain \$100
  - with chance $\frac{1}{3}$, gain \$200
  - with chance $\frac{1}{3}$, lose \$100
- then $\text{E[wealth]} = \frac{1}{3}(300) + \frac{1}{3}(400) + \frac{1}{3}(100)$
- and $\text{EU} = \text{E[utility of wealth]} = \frac{1}{3}\sqrt{300} + \frac{1}{3}\sqrt{400} + \frac{1}{3}\sqrt{100}$
- coming up later today: the shape of u entirely captures your feelings about *risk*

Outline for Rest of Today

- first, the formal assumptions we need to make on preferences over lotteries, in order to get an expected utility representation
- then, we'll specialize to the case where all outcomes involve monetary prizes, in order to link up preferences over wealth with attitudes towards risk

Terminology

- a *lottery* on a set $X = \{x_1, x_2, ..., x_n\}$ of prizes is a probability distribution $(p_1, p_2, ..., p_n)$, where p_i is the probability of prize x_i
- let P be the set of all lotteries on X: that is,

$$P = \{p = (p_1, ..., p_N) \in \mathbb{R}^N_+ \mid p_1 + .. + p_N = 1 \text{ and } p_i \geq 0 \ \forall i\}$$

Assumption: Consequentialism

- Assume: all the DM cares about is the lottery over final outcomes
- we will thus say that two lotteries with the same *reduced form* are equivalent
- for example, the following two lotteries are equivalent:
  - Lottery 1: get $3000 with chance 0.5, $0 with chance 0.5
  - Lottery 2: First, a coin is flipped, and you get lottery p^1 if it comes up heads, p^2 if it comes up tails. Lottery p^1 pays $3000 for sure. Lottery p^2 pays $0 for sure.

Assumption: Consequentialism

- here's an example where it may feel a tiny bit less innocuous to some
- Lottery 1: get \$3000 w.p. 0.5, and \$0 w.p. 0.5
- Lottery 2: I will give you \$3000. Then, force you to play a game where:
  - w.p. 0.5, you keep the money
  - w.p. 0.5, I take it back from you

Assumptions: Preferences over Lotteries

- once we've assumed consequentialism, it is safe to just talk about preferences over lotteries
  - i.e., they don't depend on how the lottery is described
- so, consider a binary relation $\succsim$ on P
- and consider making the following assumptions:

A.1 $\succsim$ is a preference relation (i.e. complete and transitive)

A.2 $\succsim$ is *continuous:* For any $p, q, r \in P$, if $p \succ q \succ r$, then there exists $\alpha, \beta \in (0, 1)$ such that:

$$\alpha p + (1 - \alpha) r \succ q \succ \beta p + (1 - \beta) r$$

Continuity

- intuition: if $p \succ q$, then sufficiently small adjustments to p (i.e. mixing it with a sufficiently small chance of "bad" lottery r) will not reverse the preference
- and similarly applied to $q \succ r$, then small adjustments to r (i.e. mixing it with a sufficiently small chance of "good" lottery p) will not reverse the preference
- If "beautiful uneventful trip by car" is preferred to "stay at home", then mixing the beautiful uneventful trip with a sufficiently small chance of "death by car accident" still results in a lottery that beats staying at home
- this rules out lexicographic preferences (i.e. "safety first")

Representations

- just as in Lecture 1, these assumptions are enough to say that preferences can be represented by *some* utility function
- but for practical purposes, this doesn't give us enough structure
- to get a more useful *expected utility* representation, we need to make one more very important assumption:

Independence Axiom

A.3. For any $p, q, r \in P$ and $\alpha \in (0,1)$:

$$p \succ q \Rightarrow \alpha p + (1-\alpha) r \succ \alpha q + (1-\alpha) r.$$

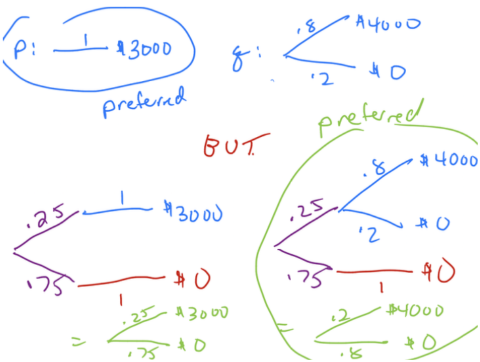
- for example: if you prefer vegetarian restaurants to Brazilian steakhouses, then you also prefer a restaurant that flips a coin in deciding whether to serve vegetarian food or pizza, to one that flips the same coin in deciding whether to serve Brazilian-style steak or pizza

Independence is a Strong Assumption

Allais Paradox

- which would you prefer: \$3000, or an 80% chance at \$4000?
- which would you prefer: a 25% chance at \$3000, or a 20% chance at \$4000?
- most prefer the certain \$3000, but the 20% chance at \$4000
- but this violates the independence axiom! (see next slide)

Independence is a Strong Assumption



EU Representation Theorem

Theorem

(von Neumann-Morgenstern) A binary relation $\succsim$ over the set P of lotteries satisfies Axioms A.1-A.3 if and only if it has an expected utility representation:

$$\text{lottery } p \succsim \text{lottery } q \text{ iff } \sum_{i=1}^{n} p_i u(x_i) \geq \sum_{i=1}^{n} q_i u(x_i)$$

Moreover, u is unique up to an affine transformation: if two functions u and v both work to represent preferences, then there exist $a > 0$ and $b \in \mathbb{R}$ such that $v = \alpha u + b$.

Expected Utility: Specialized to Money

- in the case where outcomes are monetary prizes, the representation guarantees:
  - we can find a utility function u (over money), such that your *utility* from a lottery $(p_1, p_2, ..., p_n)$ over wealths $(w_1, w_2, ..., w_n)$ is

$$EU = p_1 u(w_1) + p_2 u(w_2) + ... + p_n u(w_n)$$

- and you prefer lottery p to lottery q iff p has a higher *expected utility* than q
- the utility-of-wealth function, u, is called the vNM utility function, and we'll see that the shape of it **entirely captures attitutes towards risk**

Example

- consider a guy with initial wealth \$200, who gets utility $u(w) = \sqrt{w}$ from wealth w and faces the following gamble:
  - with chance $\frac{1}{2}$, gain \$100
  - with chance $\frac{1}{2}$, lose \$100
- calculate expected final wealth, and expected final utility

Example, Cont'd

- final wealth is $\begin{cases} 300 & \text{w.p. } \frac{1}{2} \\ 100 & \text{w.p. } \frac{1}{2} \end{cases}$
- so expected final wealth is

$$\frac{1}{2}(300) + \frac{1}{2}(100) = 200$$

- expected final utility is:

$$\frac{1}{2}\sqrt{300} + \frac{1}{2}\sqrt{100} = 13.66$$

- observe: if he got the expected final wealth as a lump sum, utility would be $\sqrt{200} = 14.142$
- this suggests that our guy **dislikes risk**

Example 2

- let's try the same exercise with $u(w) = w^2$
- here,

$$EU = \frac{1}{2}(300)^2 + \frac{1}{2}(100)^2 = 50,000$$

- if he got the expected final wealth as a lump sum, utility would be $u(200) = (200)^2 = (200)^2 = 40,000$
- so it seems that this guy **likes** risk: he would prefer to get expected final wealth \$200 via a risky gamble, rather than as a lump sum

Risk Attitudes

- our example suggested that someone with $u(w) = \sqrt{w}$ dislikes risk
- and someone with $u(w) = w^2$ likes risk
- general pattern?

Risk Aversion

Most people are *risk-averse,* meaning that they dislike risk:

- they would pay to avoid risk (so, for example, might pay for health insurance even if they don't expect to need it, in which case the cost exceeds the expected financial benefit)
- definition: a *risk-averse* DM prefers the expected monetary value of a risky gamble, to the gamble itself
- mathematically, this means

$$\underbrace{u(p_1 w_1 + ... + w_n p_n)}_{u(E[W])} > \underbrace{p_1 u(w_1) + p_2 u(w_2) + ... + p_n u(w_n)}_{E[u(W)]}$$

- this holds whenever u is a *concave function*: $u'' < 0$ (changes in wealth matter less as you become wealthier)

Other Possible Risk Attitudes

- some people are *risk-loving*: given a risky gamble, they'd rather take the gamble than get its expected monetary value as a lump sum
- that is, for a risk-lover, $u(E[W]) < E[u(W)]$, i.e.

$$u\left(p_1 w_1 + ... + p_n w_n\right) < p_1 u(w_1) + p_2 u(w_2) + .. + p_n u(w_n)$$

  Mathematically, this holds whenever u is convex, i.e. $u'' > 0$ (and $u' > 0$)
- and some people are *risk-neutral:* they just don't care about risk, and so for them, $u(E[w]) = E[u(w)]$, i.e. getting $E[w]$ for sure yields the same expected utility as playing a risky game with expected wealth $E[w]$; mathematically, this holds when u is a linear function, e.g. $u(w) = w$

Pictures

Comprehension Question

- Melsi is strictly risk-averse
- how would she rank the following gambles?
  - gamble A: definitely get \$300
  - gamble B: get \$200 w.p. $\frac{1}{2}$ and \$400 w.p. $\frac{1}{2}$
  - gamble C: get \$100 w.p. $\frac{1}{2}$ and \$400 w.p. $\frac{1}{2}$
  - gamble D: get \$200 w.p. $\frac{1}{2}$ and \$500 w.p. $\frac{1}{2}$
- [answer: $A \succ B \succ C$. D is better than B, but could be either better or worse than A]

Application 1: Insurance

- consider a risk-averse agent with initial wealth w_0, who expects to incur a loss of L with chance π
- if he can buy any level of insurance coverage q that he likes at a price of p per dollar of coverage, how much should he buy?
  - this means that he must pay a premium of pq upfront, but will get q back in the event of an accident

Example 2: Solution

- the DM wants to choose q to maximize

$$EU = (1-\pi)u(w_0 - pq) + \pi u(w_0 - pq - L + q)$$

- the F.O.C. for interior q is:

$$\underbrace{-p(1-\pi)u'(w^{\text{no accident}}) + \pi(1-p)u'(w^{\text{accident}})}_{EU'(q)} = 0$$

- you'll find that $EU''(q) < 0$ (look for an interior solution) iff the DM is risk-averse

Example 2: Interior Solution

- we can rearrange the FOC as:

$$\frac{u'\left(w^{\text{no accident}}\right)}{u'\left(w^{\text{accident}}\right)} = \frac{\pi(1-p)}{p(1-\pi)} \qquad (*)$$

Example 2: Fair Insurance

- insurance is called *fair* if $p = \pi$
- in this case, $\frac{\pi(1-p)}{p(1-\pi)} = 1$
- so (*) implies $u'(w^{\text{no accident}}) = u'(w^{\text{accident}})$, which implies (via risk-aversion)

$$w^{\text{accident}} = w^{\text{no accident}}$$

- that is, the consumer purchases *full insurance*, so that his final wealth is the same regardless of whether or not there's an accident:

$$w_0 - pq = w_0 - L + (1-p)q \Rightarrow q = L$$

Example 2: Unfair Insurance

- of course, typically, insurance is not fair, and has $p > \pi$
- then $\frac{\pi(1-p)}{p(1-\pi)} < 1$
- so (*) implies
$$u'(w^{\text{no accident}}) < u'(w^{\text{accident}})$$
- by risk-aversion, u' is decreasing, so this implies
$$w^{\text{no accident}} > w^{\text{accident}}$$
- the consumer purchases *less* than full insurance ($q < L$), leaving himself better iff if there's no accident than if there is